

Square patch antenna with circular polarization for WLAN application

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Abstract— A compact square patch antenna is studied with different feeding techniques. The proposed antenna operates at band of 2.4 GHz for the application of WLAN. Simulation results evaluated in term of return loss, bandwidth, radiation pattern, directivity, gain and axial ratio. Gain of the proposed antennas is 3.28 dB for trimmed square patch and 2.4 dB for inset feed antenna and 2.8 dB for centre feed antenna.

Keywords— Square patch antenna, WLAN, circular polarization.

I. INTRODUCTION

Development of modern wireless system requires precision system and antenna with better performance. Circular polarization antennas are one such a kind to their applications in satellite communications, Navigation systems, and mobile communications, WLAN, Wi-MAX, WPAN and RFID [1]. The advantages of circular polarization over other polarizations are immunity to multipath fading, resistant to Faraday rotation effect and antennas are not restricted in its orientation while transmission and reception. Design of antennas for wireless communication systems should have low cost, compact design and better performance such as wide bandwidth and high gain.

We implemented antennas using transmission line model approach because of its advantages as high performance with less complexity [1]. It gives more information about geometric design of antenna. In conventional antenna design we have more freedom due to distributed nature of the antenna [9]. It is just like a RLC circuit we need to optimize according to the requirement so that it is easy of all and simple to design in this method of analysis [10]. Different feed techniques are followed so that we can evaluate a better model because each technique has its own advantage over other feed techniques like strip, probe etc. As it was designed in transmission line model we have advantages in impedance matching.

The proposed antenna is designed to be operated in the 2.4 GHz (2.25 GHz – 2.57 GHz for wireless applications. The designed antennas are passive structure operates at its half wavelength mode [2]. With reference to RFIDs, Gain and Circular Polarization (CP) are very important parameters.

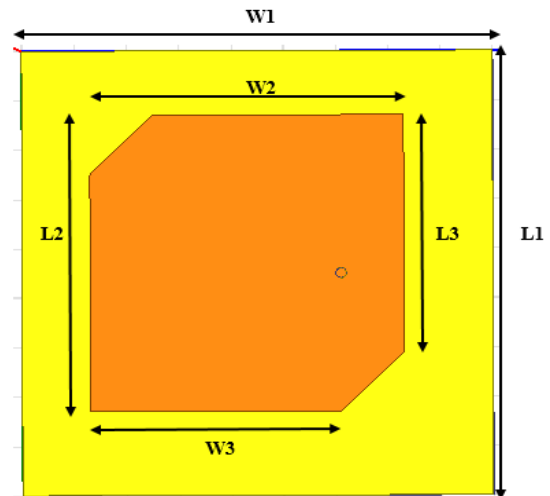
The patch elements of most commonly used shapes such as square, rectangular and circular are radiating primarily in

linear polarization [9]. To modify polarization of the patch we use conventional feed or slight changes in the patch it results circular or elliptical polarization. In conventional feed method there are no modifications in the patch but other method require changes in the patch and might be in the feed arrangements. Recently multi band antenna has greatest advantage we can design multiband antennas using different methods of implementation like slot loading, Fractal, multi stacking and modified patches. Whatever the method we choose to design multiband antenna polarization of the bands must be same so that it can be used in real time.

Gain is one of the defining factors in determining the reading range of the RFID while CP is required so that the signal can be captured effectively and successfully. Therefore, it is a need to have an antenna that has a high gain and the antenna possesses a CP instead of a linear.

II. ANTENNA TOPOLOGY AND OPERATION PRINCIPLE

The proposed antennas are as follows Fig.1. gives topology and geometry of the square patch antenna. Antenna was designed on a 1.6-mm FR4 substrate with dielectric constant $\epsilon_r=4.4$ and thickness of 0.1-mm.



(a) Trimmed square

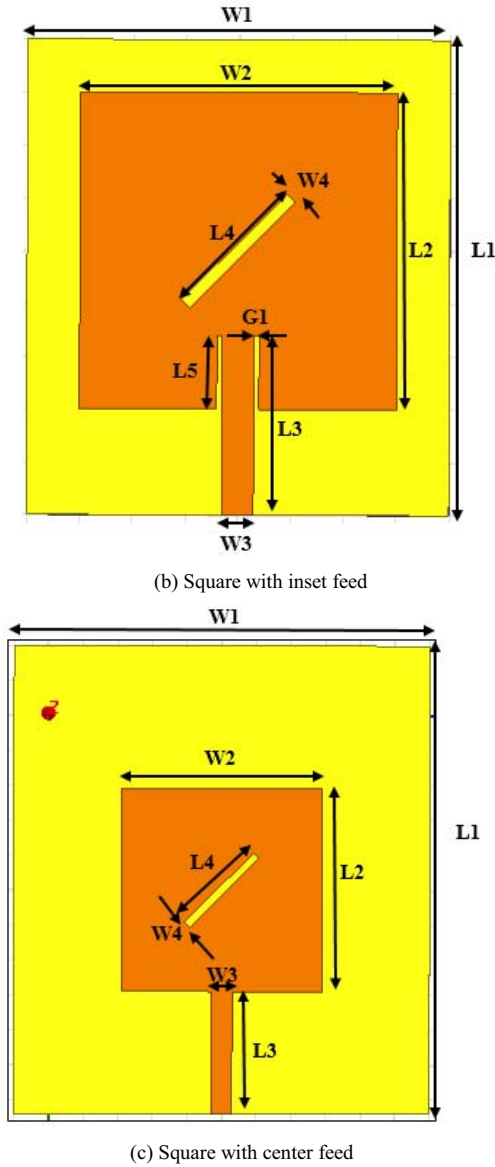


Fig.1. Topology of circularly polarized antennas

Related to our feed position it resonates at half wave length or quarter wave length or full wave length mode. However we have different methods for analysis in which transmission line model is earliest and easy than other models. It gives very good physical insight meaning to designer rather than the complex models of advanced like cavity and coupling models. Taking note point that is port 1 and verifying impedance the oscillation sustained only when Z_{ant} and Z_{in} must satisfy these conditions [9-10].

$$Z_{ant} = R_{ant} + j X_{ant} \quad (1)$$

$$Z_{in} = R_{in} + j X_{in} \quad (2)$$

$$R_{in} + R_{ant} = 0 \quad (3)$$

$$X_{in} + X_{ant} = 0 \quad (4)$$

Fig.1 (a) shows antenna structure and the geometry of a trimmed square patch. The antenna is

designed on an FR4 substrate (relative dielectric constant = 4.4, thickness = 1.6 mm, loss tan-gent = 0.025). The overall dimension of the antenna is $45 \times 45 \times 1.6 \text{ mm}^3$.

The dimensions of the antenna are: $L_1 = 45 \text{ mm}$, $L_2 = 30 \text{ mm}$, $L_3 = 25 \text{ mm}$, $W_1 = L_1$, $W_2 = L_2$, $W_3 = L_3$.

Fig.1 (b) shows antenna structure and the geometry of an inset feed square patch. The antenna is designed on an FR4 substrate (relative dielectric constant = 4.4, thickness = 1.6 mm, loss tan-gent = 0.025). The overall dimension of the antenna is $45 \times 40 \times 1.6 \text{ mm}^3$.

The dimensions of the antenna are: $L_1 = 45 \text{ mm}$, $L_2 = 30 \text{ mm}$, $L_3 = 17 \text{ mm}$, $L_4 = 14 \text{ mm}$, $L_5 = 7 \text{ mm}$, $W_1 = 40 \text{ mm}$, $W_2 = 30 \text{ mm}$, $W_3 = 3 \text{ mm}$, $W_4 = 1.2 \text{ mm}$, $G_1 = 0.5 \text{ mm}$.

Fig.1 (c) shows antenna structure and the geometry of a center feed square patch. The antenna is designed on an FR4 substrate (relative dielectric constant = 4.4, thickness = 1.6 mm, loss tan-gent = 0.025). The overall dimension of the antenna is $67 \times 60 \times 1.6 \text{ mm}^3$.

The dimensions of the antenna are: $L_1 = 67 \text{ mm}$, $L_2 = 30 \text{ mm}$, $L_3 = 17 \text{ mm}$, $L_4 = 14 \text{ mm}$, $W_1 = 60 \text{ mm}$, $W_2 = 30 \text{ mm}$, $W_3 = 3 \text{ mm}$, $W_4 = 1.2 \text{ mm}$.

III. SIMULATION RESULTS

The proposed antennas are simulated using the model simulator in HFSS. Fig.2. gives the flow analysis in HFSS. First we design the patch with proper feed and assigning the necessary boundary conditions, excitation, sweep assignment to ready with simulation.

After simulation completed we plot the results of S-parameter of two port network frequency vs. return loss. From the results obtained by S-parameter we plot other necessary antenna parameter.

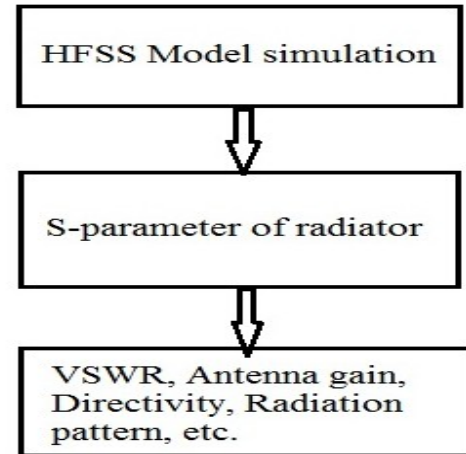


Fig. 2 Simulation flow using HFSS.

Fig.3. (a) and (b) shows the simulated results of S-parameter, axial ratio plot with respect to frequency. S-parameter result states that the proposed antenna resonates at what frequency with a return loss and bandwidth. Axial ratio gives the information about the polarization whether it is a

circular or elliptical polarization. For elliptical polarization axial ratio should be in the range of 0 to 3dB otherwise it is elliptical polarization.

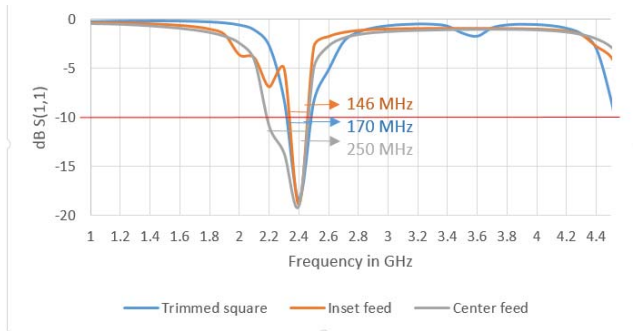


Fig. 3 (a) S-parameters of antenna frequency vs. return loss

Fig. 3 (a) shows the proposed antenna resonate at 2.4 GHz with a return loss of -19.3dB.

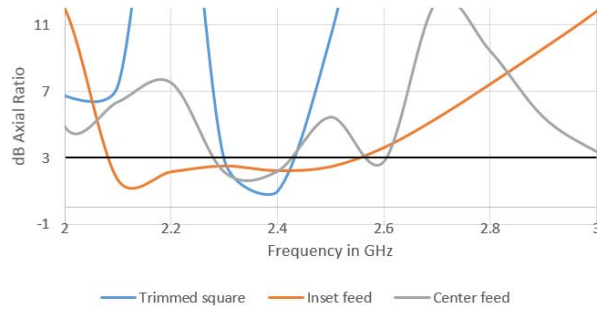


Fig. 3 (b) Axial ratio with respect to frequency

Fig.3. (b) plotted here gives the Axial ratio at resonating frequency 2.4 GHz are less than 3 this shows that the oscillations produced by the antenna under limitations to form circular polarization. It is necessary to verify with specified standards of IEEE to validate the design.

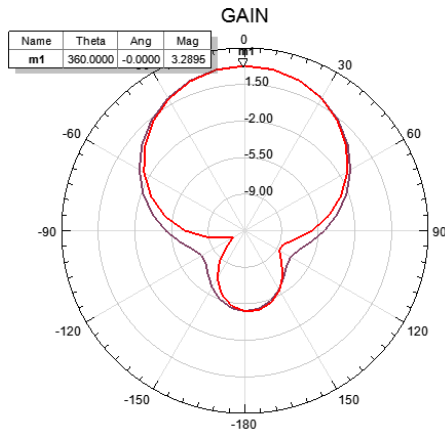


Fig. 4 (a) Radiation pattern at 2.45GHz

From the Fig.4. (a) we analyse the performance of the trimmed square antenna from its results gain, directivity and radiated power along the z direction are 3.28 dB, 6.1 dB and 20 dB respectively when , $\phi = 0^\circ, 90^\circ$ and $\Theta = 0^\circ-360^\circ$.

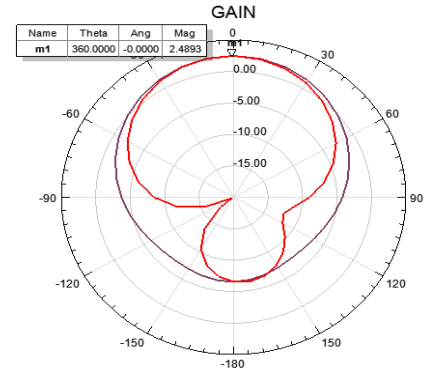


Fig. 4 (b) Radiation pattern at 2.45GHz for inset feed

From Fig.4 (b) we analyse the performance of the inset feed antenna from its results gain, directivity and radiated power along the z direction are 2.48 dB, 5.8 dB and 17 dB respectively when , $\phi = 0^\circ, 90^\circ$ and $\Theta = 0^\circ-360^\circ$.

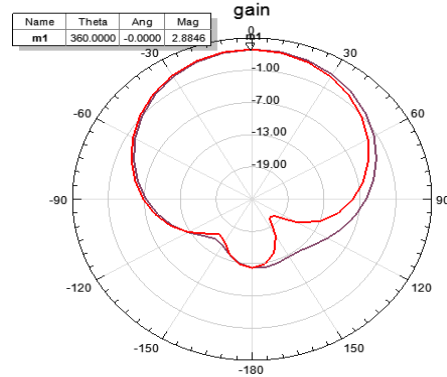


Fig. 4 (c) Radiation Pattern at 2.45GHz of centre feed

From the Fig.4 (c) we analyse the performance of the centre feed antenna from its results gain, directivity and radiated power along the z direction are 2.8 dB, 6.7 dB and 19.7 dB respectively when , $\phi = 0^\circ, 90^\circ$ and $\Theta = 0^\circ-360^\circ$.

TABLE I COMPARISONS OF ANTENNA PARAMETERS

Antenna	Trimmed square	Right-hand edge feed	Right-hand inset feed
Parameters			
Return loss in dB	-18.9	-19.6	-19
Bandwidth in MHz	170	250	146
Gain in dB	3.28	2.88	2.48
Axial ratio in dB	1.19	2.0	2.1
Size in mm ³	45x45x1.6	67x60x1.6	45x40x1.6

IV. CONCLUSION

The proposed antenna is valid for wireless communication systems and suitable candidate for other microwave applications. Proposed designs resonate at 2.4GHz frequency which has applications of WLAN. From comparison table.1 we can conclude that for gain and compact size trimmed square gives better results 3.28 dB gain with size $45 \times 45 \times 1.6 \text{ mm}^3$. For the gain and bandwidth centre feed antenna gives better results 2.8 dB, 250 MHz respectively. Trimmed square patch and other antennas can be further enhanced using wide band technique and can be used with integrating active components.

REFERENCES

- [1] Bhushan Bhimrao Dhengale, Deepak C. Karia "A high Gain 2-Element Microstrip Array Antenna with Circular Polarization for RFID Applications" *IEEE International Conference on Advances in Computing, Communications and Informatics (ICACCI)* 2015.
- [2] Nagendra Kushwaha, Raj Kumar "Compact co-planar waveguide-fed Wide band circular polarized antenna for navigation and wireless applications" *IET Microwaves, Antennas & Propagation* 2014.
- [3] Cheng-Hsun Wu and Tzyh-Ghuang Ma "Miniaturized Self Oscillating Active Integrated Antenna with Quasi-Isotropic Radiation" *IEEE Transactions on antennas and Propagation*, vol. 62, no. 2, February 2014..
- [4] Zhe Chen, Wei Hong, Ji Xin Chen, Jianyi Zhou, Lin Sheng Li "Low phase noise oscillator utilizing high-Q active resonator based on substrate Integrated waveguide technique" *IET Microwaves, Antennas & Propagation* doi: 10.1049/iet-map.2013.0380.
- [5] J.-C. Liu, C.-S. Cheng, H.-C. Chen and P.-C. Chen "Active integration ring antenna/phase shifter for direct conversions" *IEE Proc.-Microwave Antennas Propagation*. Vol.151, No.4, August 2004.
- [6] Binod Kumar Kanaujia and Babau R. Vishvakarma "Analysis of Gunn Integrated Angular Ring Microstrip Antenna" *IEEE transactions on Antennas and Propagation*, vol. 52, no. 1, January 2004.
- [7] Cheng-Hsun Wu and Tzyh-Ghuang Ma, "Miniaturized Self Oscillating Active Integrated Antenna with Quasi - Isotropic Radiation," *IEEE transactions on Antennas and Propagation*, vol. 62, no 2, February 2014.
- [8] Microwave_Engineering_David_M_Pozar 4ed Wiley 2012.
- [9] Antenna Theory Analysis and Design Constantine A_Balanis.
- [10] Mouhouche Faiza, "Analysis of small micro-strip antenna for mobile Communications", *Hand book for micro-strip antenna* September.2011